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Activity - 5

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1. 4371, 1323, 6173, 4199, 4344, 9679, 1989 &

$$h(x) = x \cdot 10$$

a) separate chaining

$$h(4371) = 1$$

$$h(1323) = 3$$

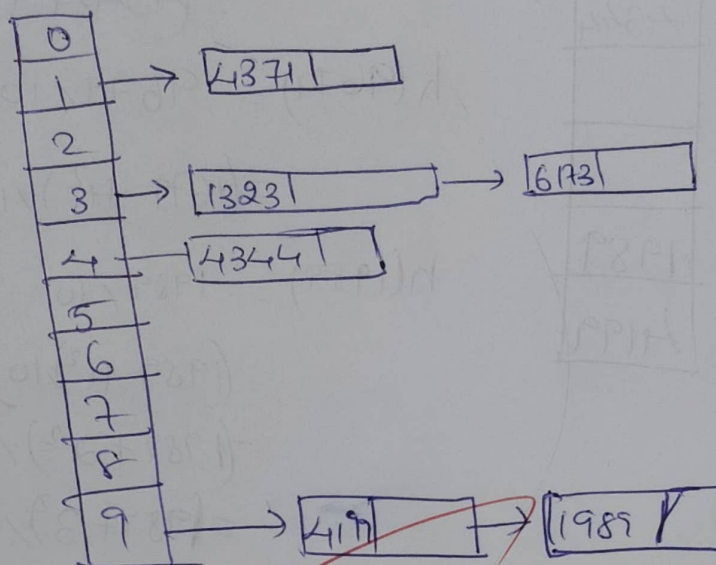
$$h(6173) = 3$$

$$h(4199) = 9$$

$$h(4344) = 4$$

$$h(9679) = 9$$

$$h(1989) = 9$$



b) linear probing

0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	
7	
8	
9	4199

$$h(6173) = 6173 \cdot 10 = 3 \text{ (collision)}$$

$$= (6173 + 1) \cdot 10 = 4$$

$$h(4344) = 4344 + 1 \cdot 10 = 5$$

$$h(9679) = 9679 + 1 \cdot 10 = 0$$

$$h(1989) = 1989 + 1 \cdot 10 = 0$$

$$1989 + 2 \cdot 10 = 1$$

$$1989 + 3 \cdot 10 = 2$$

c) Quadratic probing:

0	9679
1	4371
2	
3	1323
4	6173
5	4344
6	
7	
8	1989
9	4199

$$h(6173) = 6173 \times 10 = 3 \text{ (collision)}$$

$$= 6173 + 1^2 \times 10 = 6174 \times 10 = \textcircled{4}$$

$$h(4344) = 4344 \times 10 = 4 \text{ (collision)}$$

$$= (4344 + 1^2) \times 10 = \textcircled{5}$$

$$h(9679) = 9679 \times 10 = 9 \text{ (collision)}$$

$$= (9679 + 1^2) \times 10 = 9680 \times 10 = \textcircled{0}$$

$$h(1989) = 1989 \times 10 = 9 \text{ (collision)}$$

$$= (1989 + 1^2) \times 10 = 0$$

$$= (1989 + 2^2) \times 10 = 3$$

$$= (1989 + 3^2) \times 10 = \textcircled{8}$$

d) Double probing:

0	
1	4371
2	
3	1323
4	6173
5	9679
6	
7	4344
8	
9	4199

$$h(6173) = (6173 \times 10) + (7 - 6173 \times 7) \times 1 = \textcircled{4}$$

$$h(4344) = (4344 \times 10) + (7 - 4344 \times 7) \times 1 = 9$$

$$= (4344 \times 10) + (7 - 4344 \times 7) \times 2 = \textcircled{7}$$

$$h(9679) = (9679 \times 10) + (7 - 9679 \times 7) \times 1 = 1$$

$$= (9679 \times 10) + (7 - 9679 \times 7) \times 2 = 3$$

$$= (9679 \times 10) + (7 - 9679 \times 7) \times 3 = \textcircled{5}$$

1989 $\Rightarrow$ Does not exist.